tirvi | INITIATIVE BRIEF

HEBREW · DYSLEXIA · ACCOMMODATIONS · ISRAEL

Hebrew exam reading, *redesigned.*

A reader for dyslexic Israeli students that's not human — but reads like one. The case for tirvi: the gap, the audience, and the structure that closes it.

AUTHOR
TIRVI RESEARCH

PUBLISHED
MAY 2026

STATUS
V1.0 · CURRENT

DESIGNED WITH JISHU · JISHUTECH.IO

CONTENTS

Inside this brief.

Seven sections, ten pages. The problem, the gap, the moat, and the partnership that makes it real.

01 p. 03 — The Problem
Why generic Hebrew TTS breaks on exam content.

02 p. 04 — Why Hebrew Is Hard
No vowels, heavy acronyms, RTL meets LTR.

03 p. 05 — Market Size
The accommodation cohort, by the numbers.

04 p. 06 — The Gap
Generic readers, generic results — a comparison.

05 p. 07 — The Solution
Don't add TTS to OCR. Add a brain.

06 p. 08 — The Moat
Five micro-stages between text and audio.

07 p. 09 — Prototype & Path
Where we are, what comes next.

01 / PROBLEM

Hebrew exam reading is broken.

Israeli students with dyslexia are entitled to a human reader during exams. When the human is replaced by software, today's software fails them — not because Hebrew TTS is bad, but because no one wired Hebrew NLP into a reader that understands an exam.

Three reasons today's tools don't work

- **Pronunciation guesses.** Hebrew is written without vowels. Generic TTS guesses, gets homographs wrong, and a dyslexic ear can't recover from the error.
- **Flat reading.** Question stems, answer choices, table cells, math — all read top-to-bottom as one stream. A human reader re-reads the choices on demand. Software doesn't.
- **No accommodation-grade UX.** No replay-this-sentence, no slow-this-word, no jump-to-question, no word-sync highlight. Public criticism of Israel's 2024 computerized-reading pilot says it directly.

> *"Cold, rigid, not adjustable in real time."*
>
> PUBLIC CRITICISM — ISRAELI MOE 2024 COMPUTERIZED-READING PILOT FOR BAGRUT MATH, REPLACING HUMAN READERS FOR STUDENTS WITH DYSCALCULIA / DYSGRAPHIA. YNET, 2024.

54%

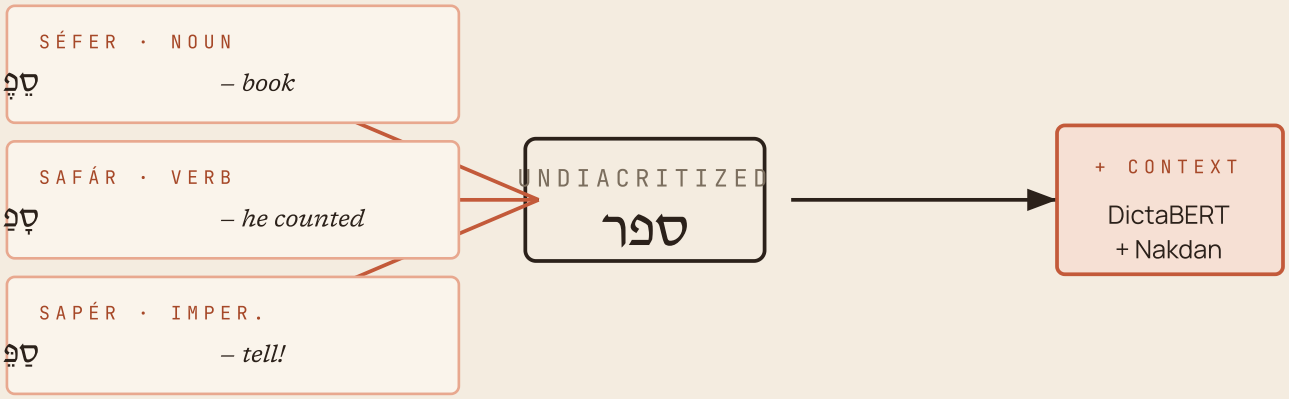
of Israeli high-schoolers received Bagrut accommodations in 2021 — up from 35% in 2011. Level-1 accommodations alone jumped 8% → 40% (2016–2021).

TAUB CENTER, 2024 · BAGRUT EXAM ACCOMMODATIONS

01.1 / WHY HEBREW IS HARD

Three things English readers never face.

English TTS works because English on the page already encodes the pronunciation. Hebrew doesn't. The reader has to think before it can speak.



01 · NO VOWELS

Same letters, three readings.

Hebrew omits ניקוד. ספר is book / counted / tell. Only morphology + context can choose. Generic TTS picks one and ships it.

02 · ACRONYMS

Heavy density, idiomatic expansion.

ד״ר → doctor, עמ׳ → page, מס׳ → number. Generic TTS spells them letter-by-letter. Hebrew exam content is full of them.

03 · MIXED DIRECTION

RTL meets LTR, mid-sentence.

Bagrut math and English mix Hebrew with English, numbers, formulas. Google's SSML `<lang>` tag returns silence on Hebrew. One sentence, three scripts, no clean handover.

02 / MARKET SIZE

The accommodation cohort, by the numbers.

Israel formally classifies more dyslexic students than almost any peer system, and the share is still rising.

~500K

Israeli students estimated to have dyslexia — about three to five in every class of thirty.

ISRAEL NATIONAL NEWS

54%

of Israeli high-schoolers received Bagrut accommodations in 2021 — vs. ~15% international LD prevalence.

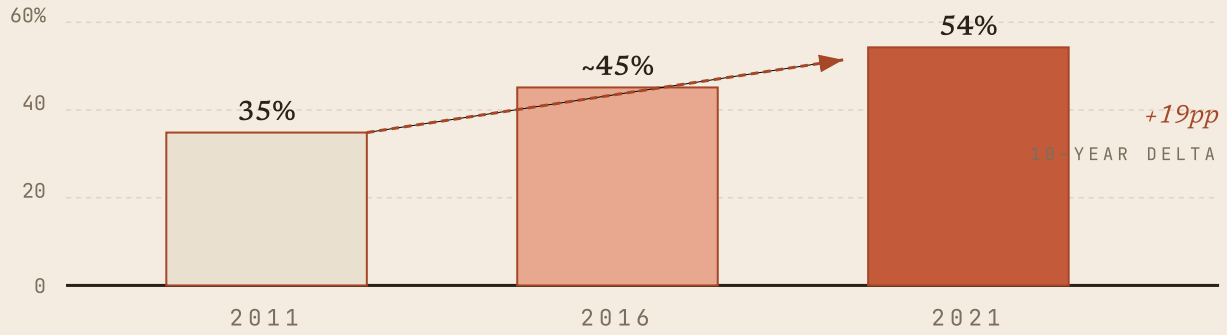
TAUB CENTER · 2024

+19pp

growth in the accommodation rate over a decade (2011 → 2021), driven by Level-1 accommodations.

TAUB CENTER · 2024

Bagrut accommodations rate, Israeli high-schoolers



SOURCE · TAUB CENTER 2024 · BAGRUT EXAM ACCOMMODATIONS

02.1 / EXISTING TOOLS

Generic readers, generic results.

No commercial product today combines Hebrew OCR, Hebrew NLP, and Hebrew TTS into an exam-shaped reader. The building blocks exist. No one has wired them together.

TOOL	HEBREW TTS	EXAM STRUCTURE	DIACRITIZATION	WORD-SYNC	VERDICT
ElevenLabs (v3, Flash v2.5)	●	○	○	●	Generic API
Speechify	●	○	○	●	Generic article reader
NaturalReader	●	○	○	◐	Generic doc TTS
Voice Dream Scanner	○	◐	○	●	Hebrew not supported
Kurzweil 3000	○	●	○	●	No Hebrew voice
MS Immersive Reader	◐	◐	○	◐	RTL layout issues
Read&Write (Texthelp)	○	◐	○	●	No first-class Hebrew
Israeli MoE 2024 pilot	●	○	◐	○	“Cold, rigid, not adjustable”
tirvi	●	◐	●	●	Built for Hebrew exams

● FULL SUPPORT ◐ PARTIAL / LIMITED ○ NONE

The building blocks – DictaBERT, Dicta-Nakdan, Phonikud, Google he-IL TTS – all exist. No one has assembled them into a vertically integrated reader for exam content. — TIRVI RESEARCH SYNTHESIS · MAY 2026

03 / SOLUTION

Don't add TTS to OCR. *Add a brain.*

The defensible layer is the middle stage — Hebrew morphology, disambiguation, diacritization, and reading-plan shaping. OCR and TTS are commodities behind adapters.

01 · INGEST	02 · INTERPRET — THE MOAT	03 · SHAPE	04 · VOICE	05 · LISTEN
Hebrew OCR	**Hebrew NLP**	**Reading Plan**	**Hebrew TTS**	**Player**
Tesseract `heb` default; Document AI fallback for hard scans. Structured blocks + bboxes, not flat text.	DictaBERT for morphology & disambiguation. Dicta-Nakdan for ניקוד. Phonikud for stress + IPA G2P.	Per-block JSON: tokens, lemma, POS, pronunciation hints, SSML cues, structural type (question / answer / table).	Google Wavenet for word-sync today. Chirp 3 HD and Azure as design-stage alternatives. Cache by content hash for cost discipline.	Side-by-side viewer, word-sync highlight, per-block playback, repeat sentence, font-size, WCAG 2.2 AA.

The three principles that decide it

- **The reading plan is the product.** 3× as much code in the interpretation layer as in adapters.
- **Adapters return rich result objects, not bytes.** Timepoints, bboxes, confidences survive the port boundary.
- **Cache by content hash, share across users.** The single biggest cost lever and the only way the $0.02 / page target survives premium voices.

Why now

- **2024 MoE pilot** proved the policy direction (digital readers) and the UX gap (cold, rigid).
- **Dicta lineage** (Bar-Ilan) shipped DictaBERT 2.0 in 2024 — SOTA on UD-Hebrew.
- **Phonikud** (June 2025) gives real-time Hebrew G2P with stress & vocal shva — the missing layer.
- **Google Chirp 3 HD he-IL** shipped in 2025 — premium voice quality at last.

03.1 / THE MOAT — HEBREW READING PLAN

Five micro-stages between text and audio.

This is where context-aware Hebrew reading happens — morphological disambiguation, pronunciation prediction, exam-domain adaptation. The academic framing matches the engineering one.

01 · TOKENIZE

Morphology

DictaBERT-large-joint — segmentation, POS, lemma, dependency. SOTA on UD-Hebrew 2024–25.

02 · DISAMBIGUATE

Context

Sentence context picks the right reading for homographs. ספר → verb or noun?

03 · DIACRITIZE

Add ניקוד

Dicta-Nakdan adds vowels for ambiguous tokens. ~86.86% word-level accuracy.

04 · G2P

Phonemes

Phonikud (June 2025) outputs IPA with stress and vocal shva. Real-time, plug into any TTS.

05 · SHAPE

SSML

Breaks between answers, emphasis on question numbers, language switch on English spans, mark for word-sync.

Worked example — one Bagrut question stem

INPUT

ספר את הסיפור במילים שלך.

DISAMBIGUATED

סַפֵּר אֶת הַסִּיפּוּר בְּמִלִּים שֶׁלָּךְ. — *"tell the story in your own words" (verb sapér, not noun séfer)*

SSML

```
<mark name="q4-w0"/><emphasis level="moderate">סַפֵּר</emphasis>
אֶת הַסִּיפּוּר <break time="200ms"/> בְּמִלִּים שֶׁלָּךְ.
```

SPOKEN

*"sa**pér** et ha-sipúr be-milím shelákh"*

04 / PROTOTYPE

A humble prototype.
A real pipeline.

A working end-to-end prototype runs today on a sample Bagrut page — Hebrew OCR, full NLP, diacritization, G2P, and word-synchronized playback. The moat is built and tested. The product around it is the next eighteen months of work.

Three horizons of work

TODAY · LIVE

The moat, working.

Hebrew OCR with layout post-processing. DictaBERT morphology and disambiguation. Dicta-Nakdan diacritization. Phonikud IPA G2P. Reading-plan assembly. Google Wavenet TTS with per-word marks. Browser playback with synchronized word highlight.

NEXT · 16 WEEKS

The product around it.

Document AI fallback for hard scans. Premium voice alternatives. Content-hash audio cache. Full block taxonomy — questions, answers, tables, figures. Side-by-side viewer. Accessibility controls. WCAG 2.2 AA pass.

HORIZON · V2

The validation layer.

tirvi-Hebrew-Exam Benchmark v0. Blind MOS evaluation with a panel of dyslexic teen readers. First institutional pilot. DPIA, parental consent ≥14, third-party copyright attestation.

The hardest part — Hebrew context-aware reading — is solved. What remains is the slower, less glamorous work of evaluation, accessibility polish, and earning the trust of the institutions that serve dyslexic students.

TIRVI RESEARCH SYNTHESIS · MAY 2026

05 / THE PATH

Built in partnership.

tirvi is too narrow to scale through venture economics, and too engineered to live as a research paper. The structure that fits is a focused Israeli nonprofit, anchored by academic, industry, and public partners.

Three pillars of partnership

01 · RESEARCH

Academic anchor.

A Hebrew NLP lab as research partner — disambiguation accuracy, diacritization evaluation, benchmark co-design. Linguistics faculty for pronunciation and accommodation-grade quality. Computer-science and linguistics students as paid contributors.

02 · ENGINEERING

Industry partner.

Cloud and runtime sponsorship — TTS, OCR, GPU inference. AI engineering mentorship. Production deployment advisory. A co-branded social-impact program with a clear AI-talent and CSR story for the partner.

03 · REACH

Public sponsor.

A university accessibility centre, a municipality, or a foundation aligned with disability and education. The first institutional pilot — students using the tool in real practice settings, with formal evaluation.

First-year commitments sought

FROM	COMMITMENT	OUTCOME
Academic anchor	One faculty advisor, two graduate-student researchers, evaluation methodology	Peer-reviewable accuracy benchmark
Industry partner	Compute credits, runtime sponsorship, one AI engineer as technical advisor	Production-grade deployment
Public sponsor	Pilot site, 10–20 dyslexic student participants, institutional cover	Validated 6-month pilot results
tirvi initiative	The full pipeline, the engineering team, the open-source codebase	A sustained, accommodation-grade Hebrew reader